

Standards & Organizations Reference Report

CarbonTrust Blockchain Carbon Credit System

This report explains:

- The main formulas and methodologies used in the CarbonTrust project (analytical model, codebase, case study).
- Which global and Indian organizations define or follow the standards behind these formulas.
- Official links for each organization.
- How current processes of these organizations are improved by the CarbonTrust design.

1 Global Standards Followed by the Analytical Model

The analytical model and codebase follow widely accepted standards and guidance from several international bodies:

- **IPCC** – 2006 Guidelines for National Greenhouse Gas Inventories (stationary combustion and emissions calculation).
- **GHG Protocol (WRI/WBCSD)** – Corporate GHG accounting and reporting standards.
- **UNFCCC / CDM** – Baseline-and-credit methodology and 1 tonne CO₂e per credit convention.
- **ICVCM** – Core Carbon Principles for high-integrity carbon credits and data quality.
- **NIST** – FIPS 180-4 Secure Hash Standard for SHA-256.
- **OASIS MQTT** – MQTT protocol specification for secure IoT messaging.
- **OWASP** – Password Storage guidance (bcrypt).
- **Ethereum / EVM community** – Smart contract and event-log patterns for on-chain credit recording.
- **IPFS / Protocol Labs** – Content-addressed storage standard via CIDs.

1.1 Moving Average Smoothing

Formula:

$$C_{\text{avg}}(t) = \frac{1}{w} \sum_{i=0}^{w-1} C(t-i)$$

Usage: Implemented in `emissioncalculator.py` and the analytical model to smooth raw CO₂ sensor readings before anomaly detection.

Standard Reference: General signal-processing and statistical convention used in environmental monitoring and QA (US EPA QA Handbook). Link: <https://www.epa.gov/sites/default/files/2020-10/documents/qa-handbook-2020.pdf>

1.2 Z-Score Anomaly Detection

Formula:

$$Z(t) = \frac{C_{\text{avg}}(t) - \text{mean}}{\text{std_dev}}$$

Usage: Implemented in `anomalydetection.py` and the analytical model to flag abnormal deviations from historical patterns.

Standard Reference: NIST Engineering Statistics Handbook describes Z-score based outlier detection. Link: <https://www.itl.nist.gov/div898/handbook/eda/section3/eda35h.htm>

1.3 Emission Rate Calculation

Formula:

$$R(t) = (C_{\text{avg}}(t) - C_{\text{bg}}) Q \alpha$$

Usage: Implemented in `emissioncalculator.py` and the analytical model to convert excess concentration and flow into a CO₂ mass-flow rate.

Standard Reference: IPCC 2006 Guidelines for National GHG Inventories, Volume 2 Chapter 2 on Stationary Combustion describe concentration-based emission estimation. Link: https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/2_Volume2/V2_2_Ch2_Stationary_Combustion.pdf

1.4 Total Emissions via Integration

Formula:

$$E_{\text{total}} = \sum_t R(t) \Delta t$$

Usage: Implemented as discrete integration in `emissioncalculator.py` to aggregate instantaneous emission rates over time.

Standard Reference: GHG Protocol Corporate Standard describes continuous emissions monitoring and integrating emission rates over time to obtain total emissions. Link: <https://ghgprotocol.org/sites/default/files/standards/ghg-protocol-revised.pdf>

1.5 Baseline and Emission Reduction

Formula:

$$E_{\text{red}} = \max(0, E_{\text{baseline}} - E_{\text{total}})$$

Usage: Implemented in `creditissuer.py` and the analytical model to compute verified emission reductions relative to a baseline.

Standard Reference: UNFCCC CDM baseline-and-credit methodologies and GHG Protocol baseline guidance. Link: <https://cdm.unfccc.int/methodologies/index.html>

1.6 Carbon Credit Conversion

Formula:

$$\text{Credits} = \frac{E_{\text{red}}}{\beta}$$

with β typically equal to 1 tonne CO₂e per credit.

Usage: Implemented in `creditissuer.py` and the analytical model.

Standard Reference: Under CDM, Verra VCS, and India's CCTS, one carbon credit usually represents one tonne CO₂-equivalent. Link: <https://cdm.unfccc.int/methodologies/index.html>

1.7 Quality-Adjusted Credits

Formula:

$$\text{Credits}_{\text{adj}} = \text{Credits} \times q$$

Usage: Implemented in `creditissuer.py` and the analytical model, using q as a data-quality factor based on anomaly rate and completeness.

Standard Reference: ICVCM Core Carbon Principles and Assessment Framework emphasize data quality, uncertainty management, and integrity, supporting quality-based adjustment of credit eligibility or quantity. Link: <https://icvcm.org/core-carbon-principles/>

1.8 SHA-256 Report Hashing

Formula:

$$\text{report_hash} = \text{SHA256}(\text{report_json})$$

Usage: Implemented in `HashUtils.kt`, `hashutils.py`, and used in `CarbonCreditRegistry.sol` to anchor emission reports on-chain.

Standard Reference: SHA-256 is defined in NIST FIPS 180-4 Secure Hash Standard. Link: <https://csrc.nist.gov/pubs/fips/180-4/upd1/final>

1.9 Blockchain Credit Record Structure

Component: `CreditRecord` struct and `issueCredit` function in `CarbonCreditRegistry.sol`.

Usage: Immutable on-chain representation of each credit including facility ID, period ID, quality-adjusted credits, total emissions, emission reduction, report hash, IPFS CID, timestamp, etc.

Standard Reference: Ethereum/EVM smart contract paradigms and ERC-style event logging. Link: <https://ethereum.org/en/developers/docs/smart-contracts/>

1.10 IPFS Content-Addressed Storage

Component: IPFS CID representing each emission report JSON.

Usage: Implemented in `ipfsclient.py` and backend to ensure tamper-proof storage of reports.

Standard Reference: IPFS protocol specification for content addressing. Link: <https://docs.ipfs.tech/concepts/content-addressing/>

1.11 Device Authentication with bcrypt

Component: Hashing of device authentication keys using `bcrypt` (`deviceauth.py`).

Standard Reference: OWASP Password Storage Cheat Sheet recommends `bcrypt` for secure password and secret storage. Link: https://cheatsheetseries.owasp.org/cheatsheets/Password_Storage_Cheat_Sheet.html

1.12 MQTT over TLS

Component: TLS-secured MQTT communication configured in `mosquitto.conf` and used by `mqttclient.py`.

Standard Reference: OASIS MQTT specification (v3.1.1 / v5.0) describes secure IoT messaging over TLS. Link: <https://docs.oasis-open.org/mqtt/mqtt/v5.0/mqtt-v5.0.html>

2 Indian Organizations and Their References

2.1 Core Indian Organizations in the Case Study

Organization	Role in India's Carbon Credit System	Official Link
Ministry of Power (MoP)	Overall policy owner of the Carbon Credit Trading Scheme (CCTS) and the Indian Carbon Market	https://powermin.gov.in/
Ministry of Environment, Forest and Climate Change (MoEFCC)	National climate strategy; formally designates covered entities under CCTS; ensures Paris Agreement alignment	https://moef.gov.in/
Bureau of Energy Efficiency (BEE)	Administrator of CCTS; sets GEI targets, methodologies, MRV rules	https://beeindia.gov.in/
Grid Controller of India Ltd (Grid-India)	National registry that issues, holds, and cancels Carbon Credit Certificates	https://www.grid-india.in/
Central Electricity Regulatory Commission (CERC)	Regulates trading of CCCs on power exchanges	https://cercind.gov.in/
National Steering Committee for Indian Carbon Market (NSCICM)	High-level committee constituted under CCTS to steer market design and targets	https://beeindia.gov.in/show_content.php?lang=1&level=1&ls_id=116&lid=294
Central Pollution Control Board (CPCB)	National body for water, air, and noise pollution monitoring standards, relevant to industrial emission compliance	https://cpcb.nic.in/
Energy Conservation (Amendment) Act, 2022	Legal basis empowering the government to establish CCTS and issue CCCs	https://beeindia.gov.in/sites/default/files/Energy%20Conservation%20Amendment%20Act%202022.pdf
Accredited Carbon Verification Agencies (ACVAs)	Third-party verifiers empanelled by BEE to verify emission and reduction data under CCTS	https://beeindia.gov.in/view_content.php?lid=568&lang=1
India.gov.in Environment Portal	National portal aggregating links and services for environment and forestry regulation	https://www.india.gov.in/topics/environment-forest/environment
Parivesh (Environmental Clearance Portal)	Online system for submitting and monitoring environmental/forest/wildlife clearances	https://environmentclearance.nic.in/

2.2 Organizations Whose Standards the Analytical Model Follows

Summarizing, the analytical model in this project is consistent with and informed by:

- **IPCC:** Emission rate formulation and stationary combustion methodology.

- **GHG Protocol:** Overall emission quantification and corporate reporting logic.
- **UNFCCC / CDM:** Baseline-and-credit structure and 1 tCO₂e per credit convention.
- **ICVCM:** Data quality and integrity principles informing the quality factor q .
- **NIST:** SHA-256 hashing for report integrity (FIPS 180-4).
- **OASIS MQTT:** Secure IoT communication standard.
- **OWASP:** Secure storage of device auth keys via bcrypt.
- **Ethereum/EVM community:** Smart contract struct and event log design.
- **IPFS / Protocol Labs:** Content-addressed storage for emission reports.
- **BEE, MoEFCC, MoP:** Policy and MRV framework for India’s CCTS into which CarbonTrust can integrate as a digital MRV layer.

3 Current Process vs. CarbonTrust Improvements (Narrative)

3.1 Monitoring, Reporting, Verification (BEE, GHG Protocol)

Current practice: Obligated entities under CCTS monitor emissions, compile reports manually, and submit them for verification by accredited agencies. This is aligned with GHG Protocol and BEE MRV rules but is periodic, document-heavy, and exposed to human error and manipulation before audits.

CarbonTrust improvement: CarbonTrust uses IoT sensors for continuous CO₂ monitoring, applies moving-average smoothing and Z-score anomaly detection automatically, and feeds verified readings directly into emission and credit calculations, reducing manual reporting and strengthening MRV.

3.2 Registry and Credit Issuance (Grid-India, UNFCCC/CDM)

Current practice: Credits are issued into a centralized registry database (Grid-India) based on verified baseline-and-credit calculations following UNFCCC/CDM-style methodologies.

CarbonTrust improvement: CarbonTrust represents each credit as an immutable on-chain `CreditRecord` on Polygon, including facility ID, period ID, total emissions, reductions, quality factor, report hash, and IPFS CID, providing a cryptographically verifiable ledger that complements the centralized registry.

3.3 Trading, Transparency, and Audit Trail (CERC, Exchanges)

Current practice: CCC trading happens on regulated exchanges under CERC oversight, with traditional transaction records and periodic reconciliation. Buyers have limited direct visibility into underlying monitoring and data quality.

CarbonTrust improvement: The Android app’s chain visualizer and credit ledger show each on-chain issuance, underlying report CID, and quality-adjusted credits, enabling transparent and immediate due diligence for managers, auditors, and potential buyers.

3.4 Data Quality and Integrity (ICVCM, NIST, IPFS)

Current practice: Data integrity is mainly assured via auditor judgment and internal IT controls. Reports are often stored on internal servers as PDFs without strong cryptographic integrity guarantees.

CarbonTrust improvement: Each emission report is hashed with SHA-256 per NIST FIPS 180-4, and the hash plus IPFS CID are stored on-chain. This combination makes any tampering immediately detectable and supports ICVCM’s emphasis on high-integrity, verifiable credits.

3.5 Double Counting Risk (MoEFCC, Paris Agreement / Article 6)

Current practice: Avoiding double counting across domestic and international markets relies on manual reconciliation and registry governance.

CarbonTrust improvement: By assigning a deterministic on-chain ID (for example, a keccak256 of facilityId, periodId, and reportHash), the system ensures each credit is uniquely traceable, providing a strong technical foundation against double issuance for the same reduction.

4 Conclusion

The CarbonTrust analytical model and implementation:

- Use formulas and methods aligned with IPCC, GHG Protocol, UNFCCC/CDM, ICVCM, NIST, OASIS MQTT, OWASP, Ethereum, and IPFS standards.
- Fit naturally into India’s CCTS architecture led by MoP, MoEFCC, BEE, Grid-India, and CERC.
- Strengthen MRV, transparency, and integrity of carbon credits through continuous IoT monitoring, statistical anomaly detection, cryptographic hashing, IPFS-backed storage, and blockchain-based recording.

These features directly address typical loopholes in current systems such as manual data handling, time lag, limited transparency, integrity risks, and double counting.